

Common cryptographic applications

Learning objectives

- **Explore HTTPS, messaging and digital signatures**
- **Explain cryptographic protections in real tools**

An introduction to cryptographic applications

- **Cryptography enables secure communication and identity**
- **Used in browsers, messaging apps and file verification**
- **Protects privacy and integrity in digital systems**

Secure websites (HTTPS)

- **Built on Transport Layer Security (TLS)**
- **Uses certificates to verify site identity**
- **Encrypts browser-server connection**

Algorithms in HTTPS

- **RSA or ECDSA for key exchange**
- **AES for content encryption**
- **SHA 256 for message integrity**

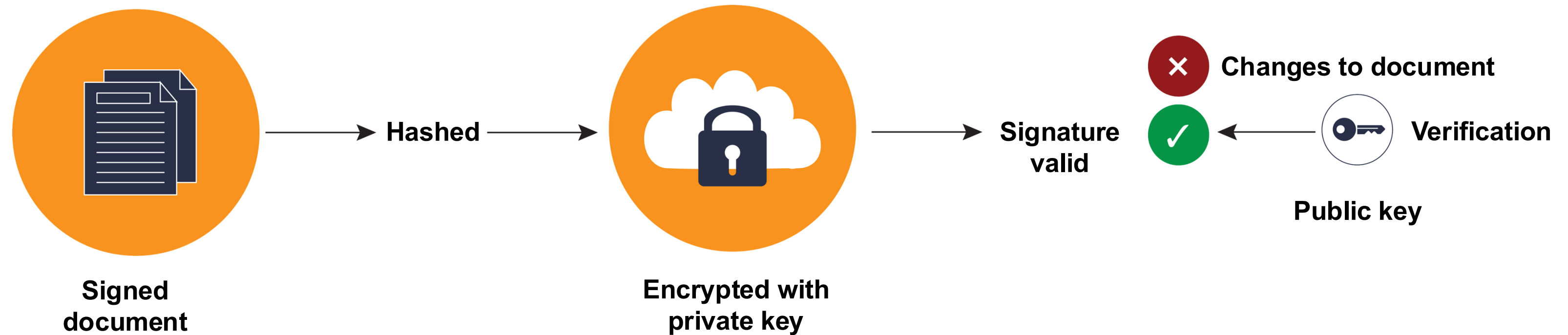
Encrypted messaging (Signal, WhatsApp)

- **End-to-end encryption protects message content**
- **Only sender and receiver can read messages**
- **Not even the app provider can access the data**

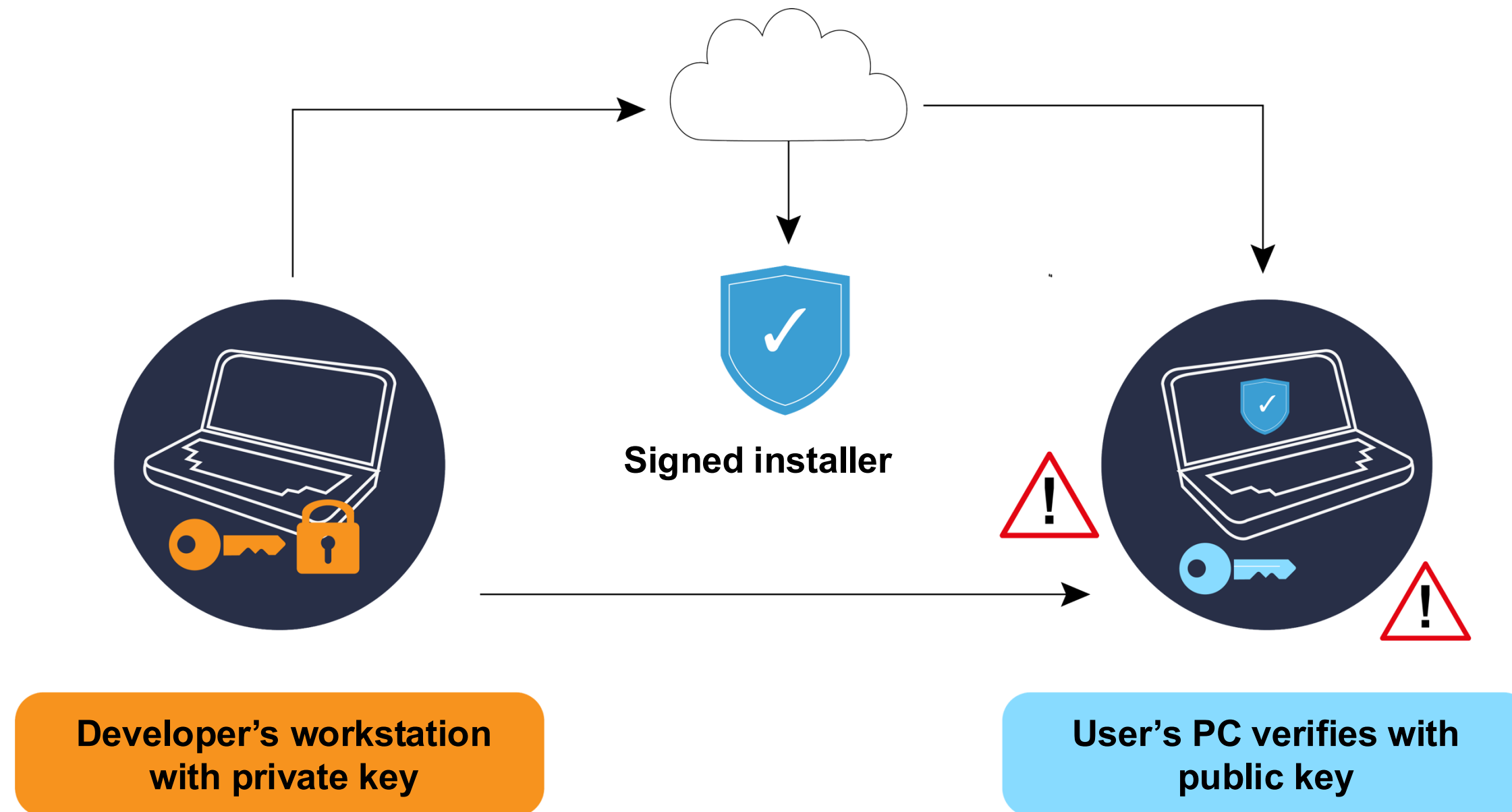
Algorithms in messaging

- **X3DH for key agreement**
- **Double Ratchet for session updates**
- **AES and HMAC for secure, verified messaging**

Digital signatures on documents



Signing software and code



Real-world examples



**Browser with
HTTPS padlock**



**WhatsApp chat
with secure icon**



**Application
download with
verified
signature**

Summary

- **Used in secure web access, communication and signature verification**
- **Cryptography ensures confidentiality, authenticity and integrity**

What's next

- **Explore brute force and dictionary attacks**